

Report: DNS & Network Security

1. Introduction

DNS and network security are fundamental components of a secure Active Directory infrastructure. Weaknesses in DNS, such as unrestricted zone transfers or unsigned records, can expose organizations to spoofing, reconnaissance, and data exfiltration attacks. This task focused on enhancing DNS security by restricting zone transfers to authorized servers, enabling DNSSEC for integrity protection, and configuring DNS logging to detect suspicious queries. These settings were applied on the Windows Server-based DNS infrastructure for the task.local domain and validated through query simulations and log analysis. These measures are vital not only for security hardening but also for compliance and incident detection in enterprise-grade environments.

2. Objective

The objective was to secure the DNS infrastructure against unauthorized zone transfers, ensure integrity of DNS responses using DNSSEC, and enable logging to detect malicious or misconfigured DNS queries. Each configuration step was meant to reduce attack surface and improve visibility over DNS operations. This contributes to the prevention of cache poisoning, data leakage, and internal reconnaissance by attackers within the network

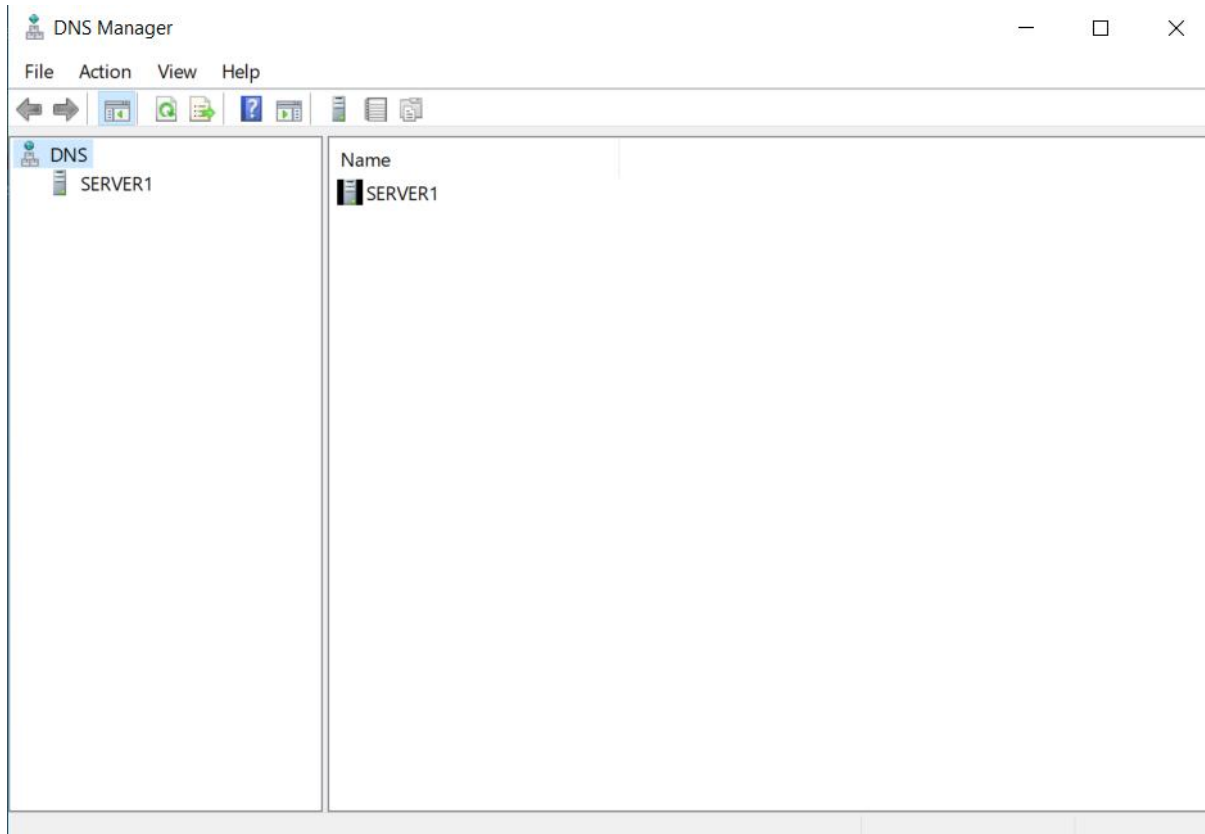
3. Tools used

- DNS Manager (dnsmgmt.msc)
- Windows Event Viewer
- PowerShell (Get-DnsServer*, Set-DnsServer* cmdlets)
- Network Monitor / Debug Logging
- Notepad (for log review)

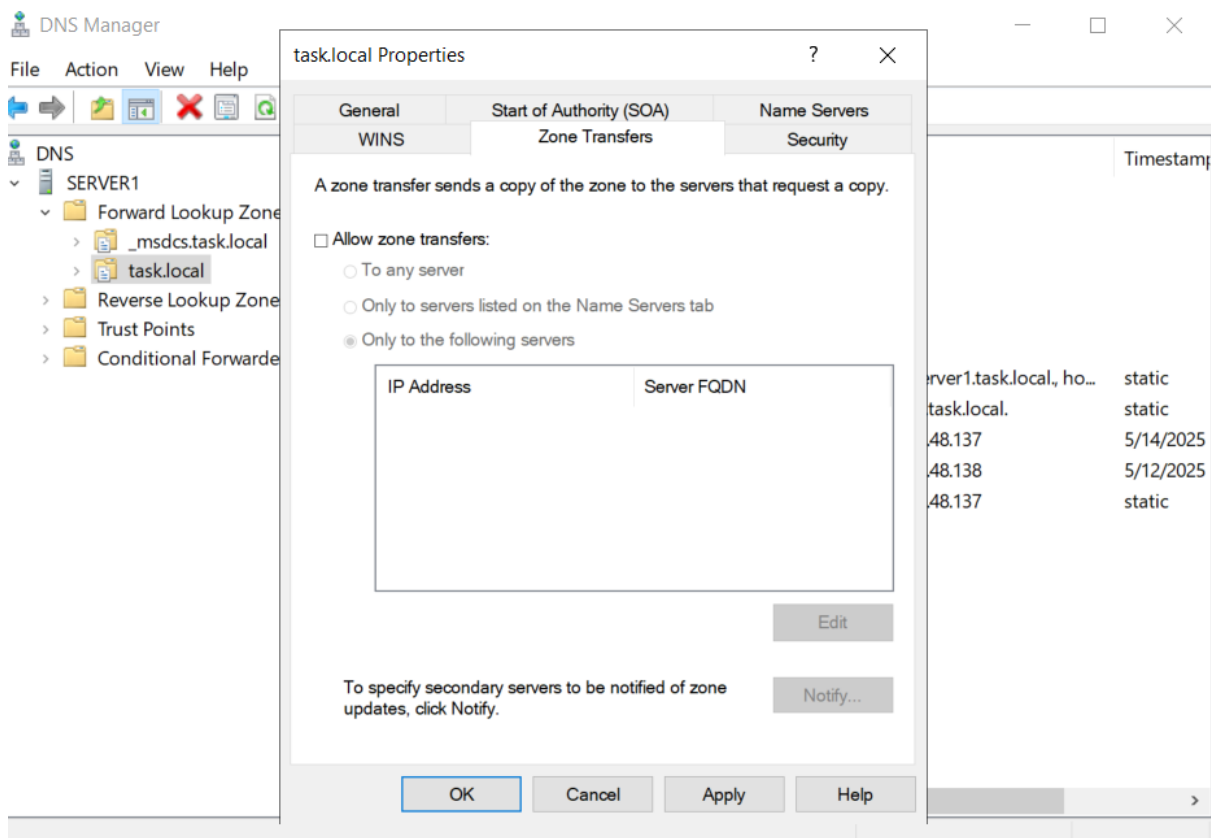
4. Methodology

4.1 Secure DNS Zone Transfers

The task.local zone was reviewed in DNS Manager on the domain controller to configure zone transfer restrictions. Since no secondary DNS server existed in the environment, zone transfers were deemed unnecessary. The zone, being Active Directory-integrated, already had zone transfers disabled by default. This was verified by accessing the zone's properties, confirming that the option to allow zone transfers was unchecked. This configuration ensured that DNS zone data could not be replicated or extracted by any unauthorized server on the network.



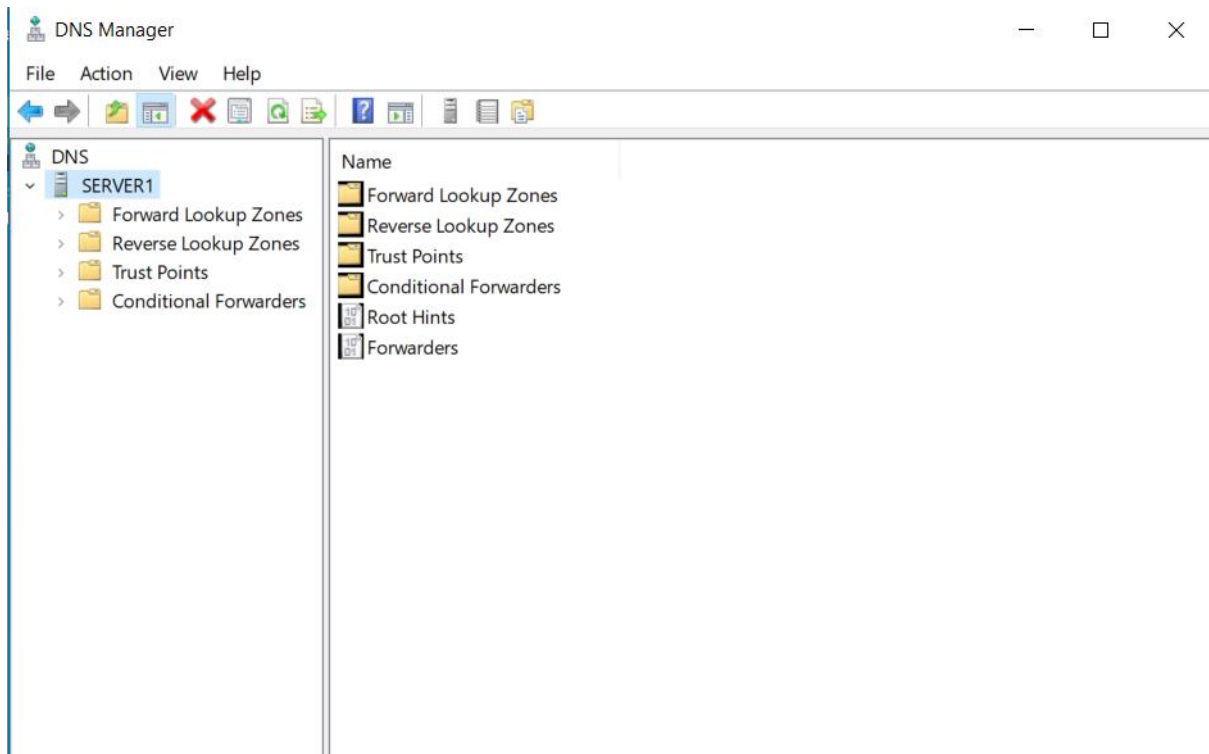
(Open DNS manager)



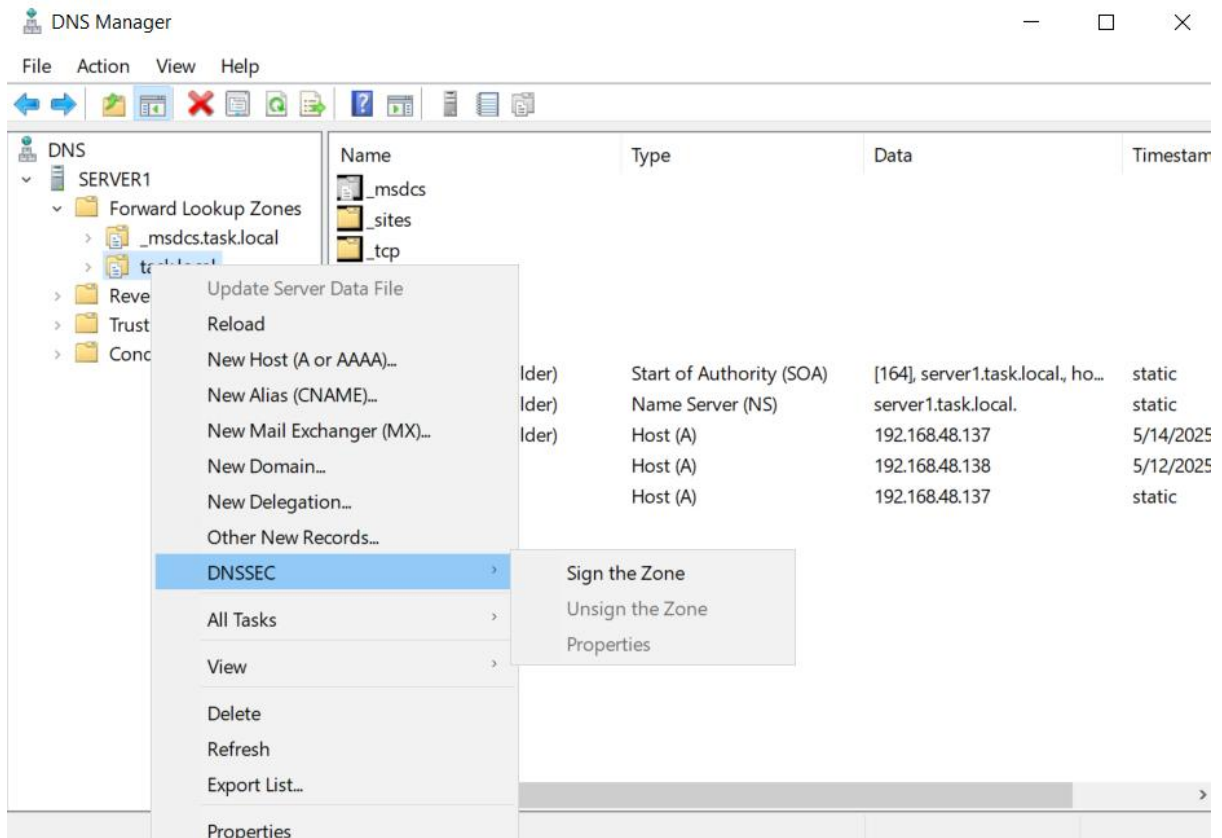
(Can't turn on zone transfer because second dns server is not available)

4.2 Enable DNSSEC

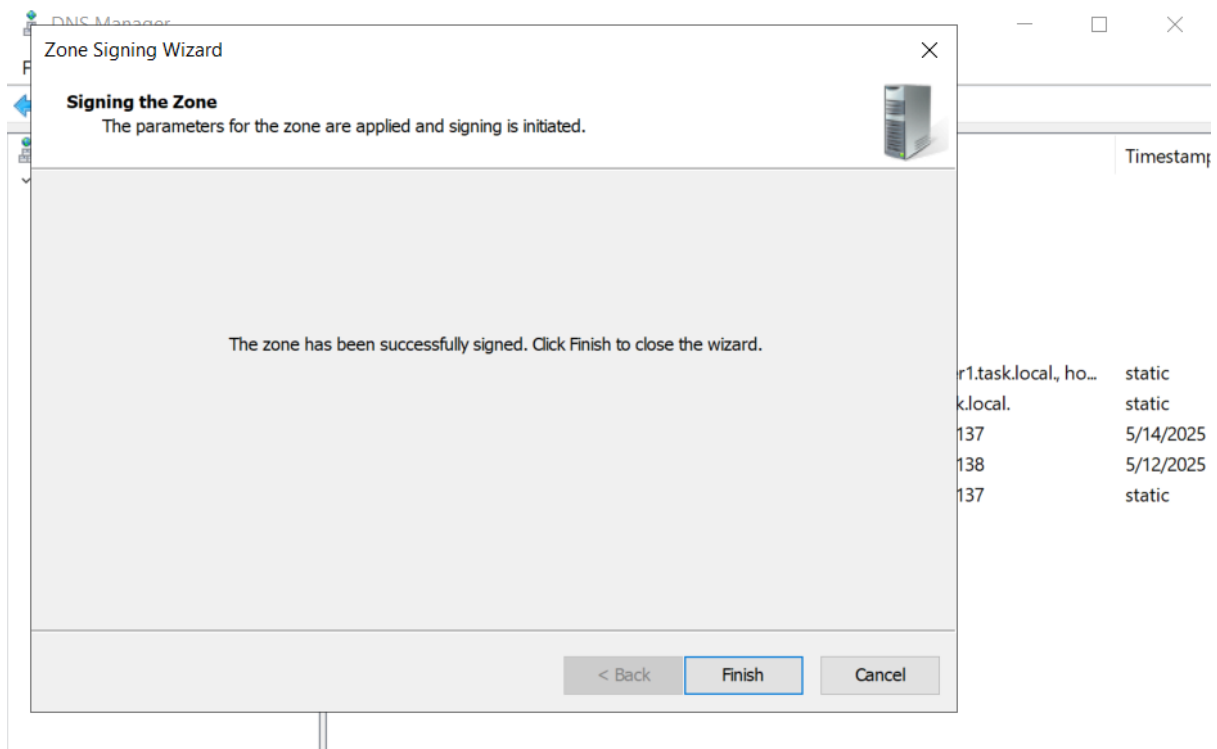
DNSSEC was enabled on the task.local zone to ensure that DNS responses could not be spoofed or altered in transit. This was done by launching the DNSSEC Signing Wizard in DNS Manager and choosing the default settings. The server automatically generated a Key Signing Key (KSK) using 2048-bit RSA and a Zone Signing Key (ZSK) using 1024-bit RSA. The algorithm used was RSA/SHA-256, with key rollover enabled and replication supported. After completion, DNSSEC-related records such as DNSKEY, RRSIG, and NSEC3 appeared in the zone, confirming that the zone was now signed and protected against forgery.



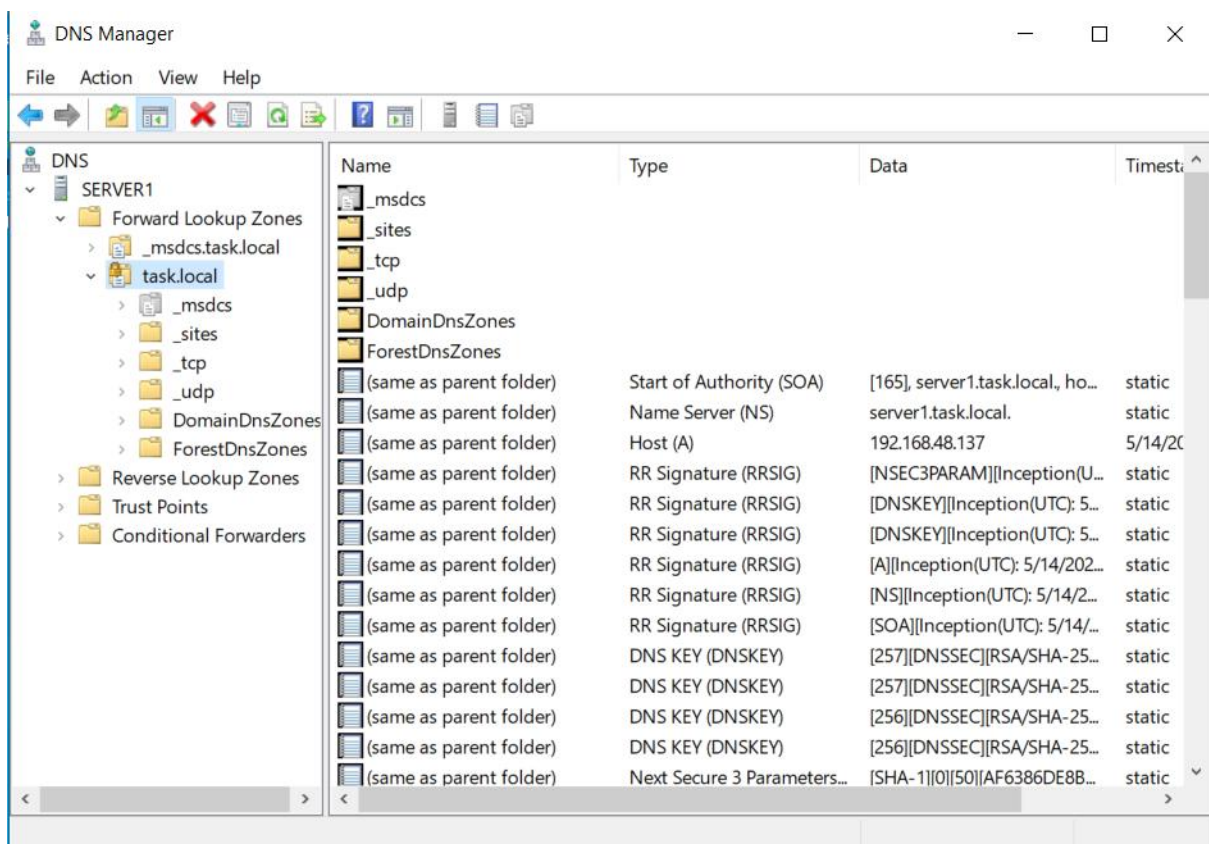
(Open DNS manager)



(Select DNSSEC and sign in)



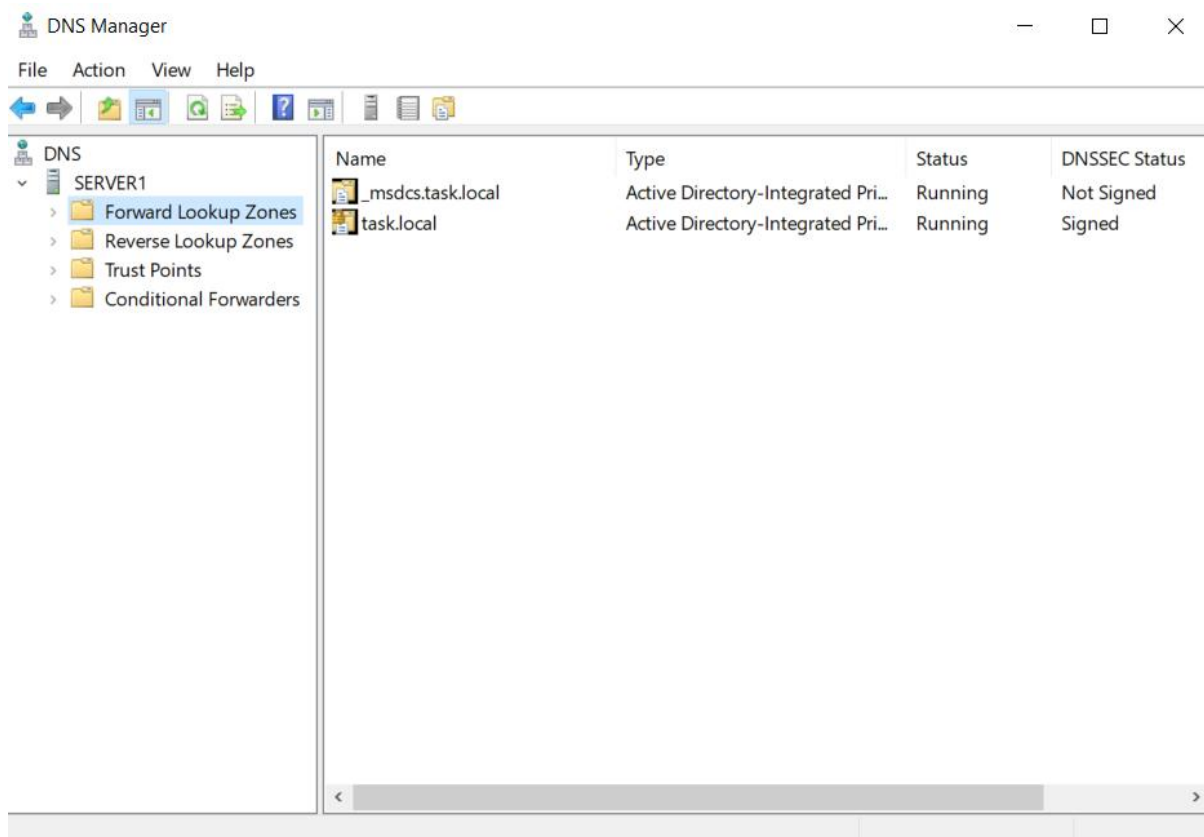
(Successfully signed in)



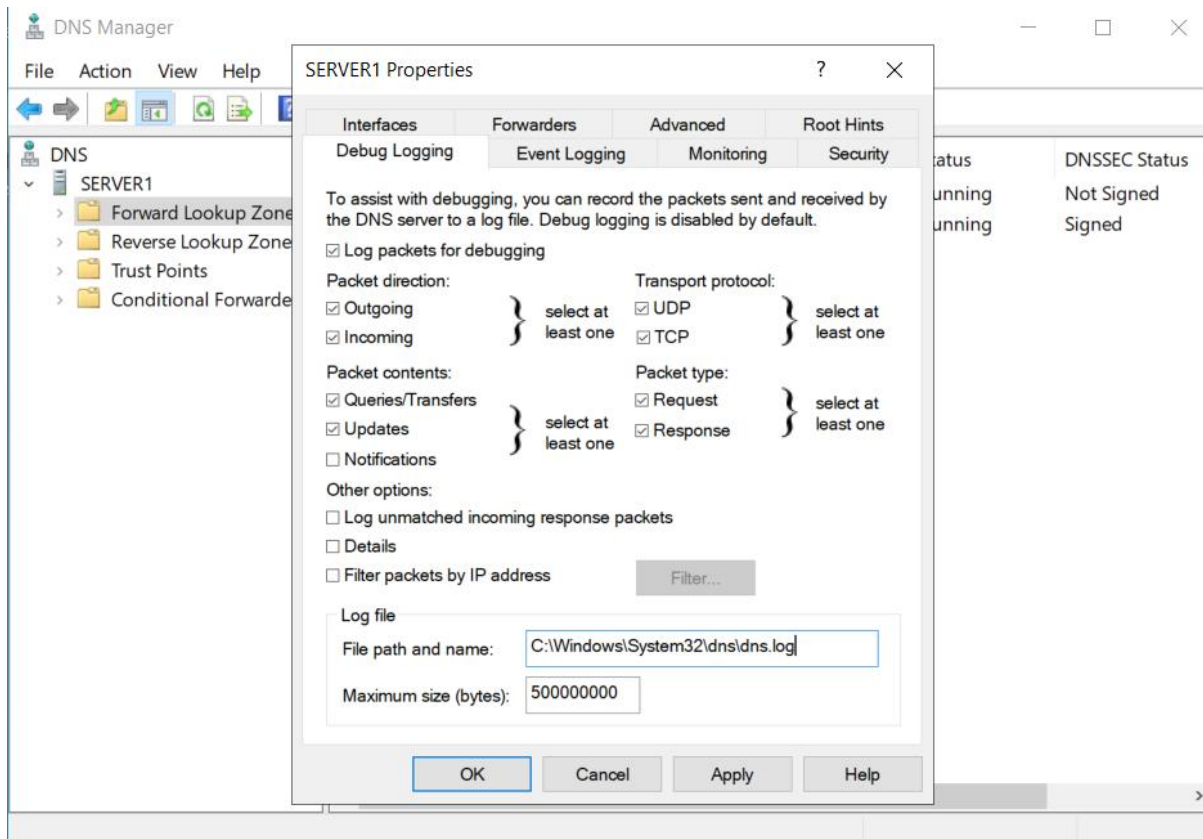
(Verified: because new folder Trust Points is added and we can see entries like RRSIG, DNSKEY in task.local)

4.3 Monitor DNS Logs for Suspicious Queries

DNS debug logging was configured through the server-level properties in DNS Manager. Logging was enabled for incoming and outgoing queries, covering both UDP and TCP traffic, as well as request and response packet types. The debug log file was set to its default location: C:\Windows\System32\dns\dns.log. After restarting the DNS service, simulated queries were run from a domain-joined client using the nslookup command. Domains such as google.com and attacker.com were queried. The resulting log entries confirmed that these queries were successfully recorded, including the full domain name, the type of query (A or AAAA), and the response codes such as NOERROR and NXDOMAIN. This demonstrated that DNS query logging was fully functional and capable of capturing both legitimate and suspicious lookup attempts.



(Open DNS manager)



(Configure SERVER1 properties)

```
PS C:\Users\Aarav.Finance> nslookup google.com
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: 192.168.48.137

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
*** Request to UnKnown timed-out
PS C:\Users\Aarav.Finance> nslookup attacker.com
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: 192.168.48.137

Non-authoritative answer:
Name:   attacker.com
Address: 209.196.146.115

PS C:\Users\Aarav.Finance>
```

(Run these commands to generate logs from PC1)

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Get-ChildItem "C:\Windows\System32\dns" | Where-Object Name -like "*.log"

Directory: C:\Windows\System32\dns

Mode                LastWriteTime         Length Name
----                -
-a-----          5/14/2025   7:59 PM           11277 dns.log

PS C:\Users\Administrator> notepad "C:\Windows\System32\dns\dns.log"
PS C:\Users\Administrator>
```

(Let's open the dns.log to check whether logs are coming or not)


```
*dns - Notepad
File Edit Format View Help
U = Update
? = Unknown

12 [ Flags (hex)
13 Flags (char codes) A = Authoritative Answer
                        T = Truncated Response
                        D = Recursion Desired
                        R = Recursion Available

14 ResponseCode ]
15 Question Type
16 Question Name

5/14/2025 7:58:27 PM 05F4 PACKET 00000237608695C0 UDP Rcv 192.168.48.138 0001 Q [0001 D NOERROR] PTR (3)137(2)48(3)168(3)192(7)in-addr(4)arpa(0)
5/14/2025 7:58:27 PM 05F4 PACKET 000002376F447560 UDP Snd 2001:7fd::1 4c3e Q [0000 NOERROR] PTR (3)137(2)48(3)168(3)192(7)in-addr(4)arpa(0)
5/14/2025 7:58:29 PM 05F4 PACKET 000002376086C080 UDP Rcv 192.168.48.138 0002 Q [0001 D NOERROR] A (6)google(3)com(4)task(5)local(0)
5/14/2025 7:58:29 PM 05F4 PACKET 000002376086C080 UDP Snd 192.168.48.138 0002 R Q [8385 A DR NXDOMAIN] A (6)google(3)com(4)task(5)local(0)
5/14/2025 7:58:29 PM 05F4 PACKET 000002376086F800 UDP Rcv 192.168.48.138 0003 Q [0001 D NOERROR] AAAA (6)google(3)com(4)task(5)local(0)
5/14/2025 7:58:29 PM 05F4 PACKET 000002376086F800 UDP Snd 192.168.48.138 0003 R Q [8385 A DR NXDOMAIN] AAAA (6)google(3)com(4)task(5)local(0)
5/14/2025 7:58:29 PM 05F4 PACKET 000002376086F9A70 UDP Rcv 192.168.48.138 0004 Q [0001 D NOERROR] A (6)google(3)com(0)
5/14/2025 7:58:29 PM 05F4 PACKET 000002376F444950 UDP Snd 2001:503:b3e::2:30 85d5 Q [0000 NOERROR] A (6)google(3)com(0)
5/14/2025 7:58:30 PM 016C PACKET 000002376F447560 UDP Snd 192.112.36.4 4c3e Q [0000 NOERROR] PTR (3)137(2)48(3)168(3)192(7)in-addr(4)arpa(0)
5/14/2025 7:58:30 PM 016C PACKET 000002376F447560 UDP Snd 192.203.230.10 4c3e Q [0000 NOERROR] PTR (3)137(2)48(3)168(3)192(7)in-addr(4)arpa(0)
5/14/2025 7:58:31 PM 05F4 PACKET 000002376F423890 UDP Rcv 192.168.48.138 0005 Q [0001 D NOERROR] AAAA (6)google(3)com(0)
5/14/2025 7:58:31 PM 05F4 PACKET 000002376F441D40 UDP Snd 2001:500:1::53 dab4 Q [0000 NOERROR] AAAA (6)google(3)com(0)
5/14/2025 7:58:31 PM 05F4 PACKET 0000023763AE3CC0 UDP Rcv 192.203.230.10 4c3e R Q [0000 NOERROR] PTR (3)137(2)48(3)168(3)192(7)in-addr(4)arpa(0)
5/14/2025 7:58:31 PM 05F4 PACKET 000002376F447560 UDP Snd 199.180.182.53 65d7 Q [0000 NOERROR] PTR (3)137(2)48(3)168(3)192(7)in-addr(4)arpa(0)
5/14/2025 7:58:31 PM 05F4 PACKET 0000023763AD6070 UDP Rcv 192.112.36.4 4c3e R Q [0000 NOERROR] PTR (3)137(2)48(3)168(3)192(7)in-addr(4)arpa(0)
5/14/2025 7:58:31 PM 05F4 PACKET 00000237653BC8E0 UDP Rcv 199.180.182.53 65d7 R Q [0000 NOERROR] PTR (3)137(2)48(3)168(3)192(7)in-addr(4)arpa(0)
5/14/2025 7:58:31 PM 05F4 PACKET 000002376F44F990 UDP Snd 2001:503:c27::2:30 046a Q [0000 NOERROR] A (1)y(4)arin(3)NET(0)
5/14/2025 7:58:32 PM 016C PACKET 000002376F444950 UDP Snd 192.112.36.4 85d5 Q [0000 NOERROR] A (6)google(3)com(0)
5/14/2025 7:58:32 PM 016C PACKET 000002376F444950 UDP Snd 192.203.230.10 85d5 Q [0000 NOERROR] A (6)google(3)com(0)
5/14/2025 7:58:32 PM 05F4 PACKET 00000237673831C0 UDP Rcv 192.203.230.10 85d5 R Q [0000 NOERROR] A (6)google(3)com(0)
5/14/2025 7:58:32 PM 05F4 PACKET 000002376F444950 UDP Snd 192.48.79.30 0f7f Q [0000 NOERROR] A (6)google(3)com(0)
5/14/2025 7:58:32 PM 05F4 PACKET 0000023767380580 UDP Rcv 192.112.36.4 85d5 R Q [0000 NOERROR] A (6)google(3)com(0)
5/14/2025 7:58:33 PM 05F4 PACKET 0000023763F24D90 UDP Rcv 192.48.79.30 0f7f R Q [0000 NOERROR] A (6)google(3)com(0)
5/14/2025 7:58:33 PM 05F4 PACKET 000002376F444950 UDP Snd 216.239.32.10 093e Q [0000 NOERROR] A (6)google(3)com(0)
5/14/2025 7:58:33 PM 05F4 PACKET 00000237620F54F0 UDP Rcv 216.239.32.10 093e R Q [0084 A NOERROR] A (6)google(3)com(0)
5/14/2025 7:58:33 PM 05F4 PACKET 000002376086F9A70 UDP Snd 192.168.48.138 0004 R Q [0001 DR NOERROR] A (6)google(3)com(0)
5/14/2025 7:58:34 PM 016C PACKET 000002376F441D40 UDP Snd 192.112.36.4 dab4 Q [0000 NOERROR] AAAA (6)google(3)com(0)
```

(Suspicious Entry Detected: Yes (attacker.com.task.local)

Standard Entry Detected: Yes (google.com))

5. Recommendations

- Keep DNS zone transfers disabled unless explicitly needed for secondary DNS replication
- Limit transfer permissions to specific IP addresses if zone transfers are required
- Leave DNSSEC enabled and monitor key rollover status periodically
- Archive and review dns.log regularly for abnormal patterns like frequent NXDOMAIN responses
- Use Event Viewer and PowerShell to cross-check audit-level logs for DNS changes
- Integrate DNS logs with a SIEM platform for proactive threat detection and alerting
- Configure client machines to support DNSSEC validation where possible

6. Conclusion

The DNS and network security configurations implemented in this task significantly hardened the internal DNS infrastructure. By disabling unnecessary zone transfers, the system was protected from unauthorized replication and information disclosure. DNSSEC was successfully implemented, providing cryptographic assurance of DNS record authenticity, which guards against spoofing and man-in-the-middle attacks. DNS debug logging was also activated, allowing real-time visibility into DNS traffic and enabling the detection of potentially suspicious activity. Through simulated queries and log inspection, the effectiveness of these measures was verified. Together, these configurations ensure that the DNS system within the task.local domain is resilient, secure, and capable of supporting forensic auditing and policy compliance.